

Visionary

BEHAVIOR recovery recipe · NO Discord/Drive

One narrative

DAgger = query the robot's mistakes. **D**ART = make the supervisor rehearse likely mistakes. **2026** = schedule those rehearsals.

49→79

steal α card

a/b/c noises

DART/FM/chunk

task-gated

ACT warning

≠ VLA child

rhyme only

no Discord

no Drive

Talk arc

BEHAVIOR needs → steal DART → three noises → follow-ups

1. BEHAVIOR dies from unrecoverable drift (slip/miss/occlusion)
2. Steal: recovery-demo protocol + α -schedule card beside $\pi_{0.5}/N1.7$
3. Separate three noises: demo DART / FM train / domain rand
4. Task-gate noise (ACT warning for fine contact)
5. Narrative: rehearse mistakes vs query mistakes
6. Motion deep dive: compounding · boundary · action $\neq$ state
7. Follow-ups: heteroscedastic $\Sigma(x)$ · DART+chunking · meta- α
8. New apps · anti-patterns · don't claim VLA descendants · tattoo

BEHAVIOR needs recovery

Long-horizon household tasks die from unrecoverable drift

Slip / miss

Gripper slip, missed grasp, object nudge — need restabilization demos.

Occlusion

Visual loss mid-horizon; policy must recover without DAgger-style danger.

Stage drift

Multi-stage BDDL tasks compound; recovery labels between stages.

Distractors

Layout churn; off-manifold states appear even with good priors.

Steal card — beside $\pi_{0.5}$ / N1.7

Recovery-demo + α -schedule — not 2017 CNN

Steal from DART

1. Optimized action noise during teleop
2. α prior / inverted-U calibration
3. Collect-time reward as ops metric
4. $K=2$ noise update after BC seed
5. Isotropic is not enough — fit Σ
6. Slogan: disturb expert, not deploy learner

Pair with 2026 stack

$\pi_{0.5}$ / N1.7 (or ACT/DP) as policy head

Chunking for horizon

DART-style demos for recovery coverage

Task-gate α : loco/nav $\uparrow$ · insert $\downarrow$

Keep-fails / intervention upsample (DROID/ π lore)

Do NOT ship paper ResNet as entry

Three noises — keep them labeled (KANTA)

(a) DART teleop · (b) FM/diffusion · (c) chunking — do not collapse

(a) DART teleop

Optimized Gaussian on
supervisor actions
during demo collection
→ recovery labels
 $\Sigma^{\alpha} + \alpha$ schedule
THIS paper · HSR $\alpha \in \{3, 6\}$

(b) FM / diffusion

Generative-head noise
for score / flow matching
Train or sample-time
 π_0 / DP / BEHAVIOR'25
 $\neq$ teleop disturbance

(c) Chunking

Predict action sequences
execute prefix · replan
ACT / DP horizon control
ACT cites DART then
prefers this for fine manip

Task-gated noise — ACT warning

Fine contact hates open noise; loco loves recovery drills

Locomotion / nav / push-clutter

Larger α OK — recovery funnel valuable

↑ noise

Coarse pick-and-place

Moderate α ; monitor collect reward

mid α

Precision insert / thread / gear

Near-zero open noise; prefer chunking

↓ / gate

Bimanual fine teleop

ACT moral: noise can cause task failure

gate hard

One narrative

Rehearse vs query — then schedule rehearsals

Dagger

Query the robot's mistakes.
Roll out learner → ask supervisor to label failures.
Costly · tedious · dangerous early.

DART

Make the supervisor rehearse likely mistakes.
Inject Σ^α into expert actions → recovery demos near the manifold.

2026

Schedule those rehearsals.
 α curriculum ·
task-gated noise ·
chunked heads ·
keep-fails ·
meta- α from priors.

Motion / covariate-shift deep dive

Required Visionary beat

Compounding over T

Household horizons make BC drift lethal — same pathology as ACT/DP chunk papers.

Recovery at boundary

Force expert to label restabilization; BEHAVIOR stage detectors can score it.

Action $\neq$ state noise

Demo DART $\neq$ FM sample noise $\neq$ domain rand — three levers, three dashboards.

Complementary to chunking

Noise for coverage; chunks for horizon; RTC/inpaint for inference repair.

Collect safety metric

Track supervisor reward during noisy teleop — -5% vs -80% is the ops story.

Follow-up research

Where to push past CoRL'17

Heteroscedastic $\Sigma(\mathbf{x})$

State-dependent noise instead of global Σ — more noise near unstable modes.

DART + chunking

Collect with gated noise; deploy chunked ACT/DP/ π — coverage $\times$ horizon.

Meta- α

Replace hand α with meta-learned target error from prior tasks / stages.

VLA co-train replay

Inject noise in proprio/action channels during replay, not only pixels.

Layered compare

CLEAR table: DART teleop vs FM sample noise vs RTC inpaint — different layers.

Keep-fails union

DROID-style retain interventions $\cup$ DART forced recoveries.

New apps

Where rehearsed recoveries pay rent

1. Assistive teleop that stays high-reward under tremor / disturbance

2. Autonomous driving lane-recovery demo collection

3. Surgical / ultrasound: safe expert disturbances beat learner-in-loop

4. Sim-to-real locomotion: persistence-of-excitation meets IL

5. BEHAVIOR household: stage-gated α between BDDL subgoals

6. Education: shortest reading for “recovery as data augmentation”

Anti-patterns → fixes

What not to do in 2026

✗ Blind isotropic on dexterous bimanual

→ Fit Σ ; task-gate α ; prefer chunking on contact

✗ Claim VLAs are DART descendants

→ Teach rhyme vs lineage; ACT is the CLEAR cite

✗ Ship 2017 CNN pushing as entry

→ Steal protocol; race on $\pi_{0.5}/N1.7$ heads

✗ Conflate FM noise with teleop noise

→ Dashboard three noises separately

✗ Ignore collect-time reward

→ Log supervisor@collect next to SR

✗ α monotone-more-is-better

→ Expect inverted-U; calibrate per operator/task

Don't claim VLAs are DART descendants

CLEAR vs TENTATIVE — Visionary must police this

HARD RULE

Modern IL/VLA robustness via diversity, chunking, keep-fails, and sample-time FM noise is mostly TENTATIVE design rhyme — not a DART citation graph. ACT is the CLEAR cite (and it rejects open noise for fine teleop).

Say

“DART-style recovery demos are a plausible BEHAVIOR data intervention.”

Don't say

“ $\pi_{0.5}$ is a DART descendant.” / “OpenVLA uses DART.”

Evidence you steal numbers from

HSR 49/79/72 · Humanoid 3x · -5% vs -80%

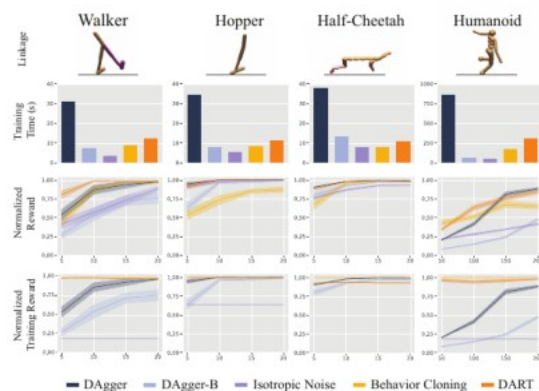


Figure 2: Top: The four different locomotive domains in MuJoCo we evaluated DART on: Walker, Hopper, Half-Cheetah and Humanoid. Top Middle: The time, in seconds, to achieve the performance level reported below. DART achieves similar performance to DAgger in all 4 domains, but requires significantly less computation because it doesn't require retraining the current robot policy after each demonstration. DAgger-B reduces the computation required by less frequently training the robot, but suffers significantly in performance in domains like Humanoid. Bottom Middle: The learning curve with respect to reward obtained during training with each algorithm plotted across the number of demonstrations. Bottom: The reward obtained during collection of demonstrations for learning. DART receives near the supervisor's reward at all iterations whereas DAgger can be substantially worse in the beginning.

policies in Tensorflow. At each iteration we collected one demonstration. In order to make the task challenging for Imitation Learning, we used the same technique as in [6], which is to sub-sample 50 state and control pairs from the demonstrated trajectories, making the learners receive less data at each iteration.

We compare the following five techniques: Behavior Cloning, where data is collected from the supervisor's distribution without noise injected; DAgger, which is an on-policy method that stochastically samples from the robot's distribution and then fully retrains the robot's policy after every demonstration is collected; DAgger-B, which is DAgger but the retraining is done intermittently to reduce computation time; Isotropic-Gaussian which injects non-optimized isotropic noise; and DART. See supplement material for how the hyper-parameters for each method are set.

We used the following MuJoCo environments: Walker, Hopper, Humanoid, and Half-Cheetah. The size of the state and control space for each task is $\{|X| = 17, |U| = 6\}$, $\{|X| = 11, |U| = 3\}$, $\{|X| = 376, |U| = 17\}$, $\{|X| = 117, |U| = 6\}$, respectively. All experiments were run on a MacBook Pro with an 2.5 GHz Intel Core i7 CPU. We measured the cumulative reward of each learned policy by rolling it for 500 timesteps, the total computation time for each method and the cumulative reward obtained during learning.

Fig. 2 shows the results. In all domains, DART achieves parity with DAgger, whereas Behavior Cloning and DAgger-B are below this performance level in Walker and Humanoid. For Humanoid, DART is 3x faster in computation time and during training only decreases the supervisor's cumulative reward by 5%, whereas DAgger executes policies that have over 80% less cumulative reward than the supervisor. The reason for this is that DAgger requires constantly updating the current robot policy and forces the robot into sub-optimal states. While, one way to reduce this computation is to decrease



Figure 3: Left: Experimental setup for the grasping in clutter task. A Toyota HSR robot uses a head-mounted RGBD camera and its arm to push obstacle objects out of the way to reach the goal object, a mustard bottle. The robot's policy for pushing objects away uses a CNN trained on images taken from the robot's PrimeSense camera, an example image from the robot's viewpoint is shown in the orange box. Right: The Success Rate for Behavior Cloning, DART($\alpha = 3$) and DART($\alpha = 6$). DART($\alpha = 3$) achieves the largest success rate.

the number of times the policy is updated. DAgger-B illustrates that this can significantly deteriorate performance. Lastly, naively applying isotropic noise does not perform well, and leads to unsafe policies during execution, which suggests the need for optimizing the level of noise.

5.2 Robotic Grasping in Clutter

We evaluate DART with human supervisors in a grasping in clutter task on a Toyota HSR robot. We consider a task inspired by the Amazon Picking Challenge [2]. The goal of the task is for the robot to retrieve a goal object from a cupboard. The task is challenging because the goal object is occluded by obstacle objects and the robot must reason about how to clear a path based on observations taken from an eye-in-hand perspective. The objects are 6 common household food items, which consist of boxes and bottles with varying textures and mass distributions. The target object is fixed to always be a mustard bottle. The robot, task and image viewpoint are shown in Fig. 3. See supplement material for additional information on the task.

We use 4 supervisors who have robotics experience but not specifically in the field of Imitation Learning and compare Behavior Cloning and DART. When performing the study, we first collect $N = 10$ demonstrations with Behavior Cloning (i.e. no noise) and then in a counter-balanced order collect $N = 30$ more demonstrations with each technique. Our experiment was within-subject, have every supervisor perform all three methods.

We only updated the noise parameter after the first 10 demonstrations (i.e. $K = 2$). The final robot policy was then trained on the total of 40 demonstrations. We consider two different choices of α : $\alpha = 3\text{Tr}(\Sigma_1)$ and $\alpha = 3\text{Tr}(\Sigma_1)$. These choices correspond to the intuition that with small datasets in high dimensional image space the robot will have significant shift from the current loss on the supervisor's distribution. Through the rest of the paper, we will write our choices as $\alpha = 3$ and $\alpha = 6$ for brevity in notation.

During policy evaluation, we measured success as 1) the robot is able to identify the goal object and 2) there being a clear path between its gripper and the object. Once these conditions are identified, as detailed in the supplement material, an open-loop motion plan is generated to execute a grasp around the target object.

In Fig. 3, we report the average performance of the three techniques across DART 20 times on different sampled initial states. DART with $\alpha = 3$ performs the best out of the three techniques, with a 79% success rate over traditional Behavior Cloning's 49% success rate. Interestingly, DART with $\alpha = 6$ performs better than Behavior Cloning, but only has a 72% success rate. This may suggest this level of noise was potentially too high for the human supervisor.

6 Conclusion

This paper considers injecting noise into the supervisor's demonstrations to minimize covariate shift. Our algorithm, DART, provides the robot with corrective examples at the boundary of the supervisor's policy. By being concentrated around the supervisor's policy, it collects demonstrations without visiting highly sub-optimal states and is easier for human supervisors tele-operating. We demonstrate it achieves parity with on-policy methods in simulated MuJoCo domains and is significantly better than traditional Behavior Cloning on a real robot. For code and more information see <http://berkeleyautomation.github.io/DART/>

Exam bite

Visionary one-liner for BEHAVIOR week

EXAM BITE

For BEHAVIOR: steal DART's recovery-demo + α -schedule protocol beside a modern chunked head ($\pi_{0.5}/N_{1.7}$). Separate demo noise / FM noise / domain rand. Task-gate α (ACT warning). Do not claim VLAs are DART descendants — rehearse likely mistakes in data, then schedule those rehearsals in 2026.

Tattoo card

HSR 49% / 79% / 72% · +62% rel · Humanoid 3× wall-clock · collect −5% (DART) vs −80% (DAgger) · α inverted-U · NO Discord,

49→79

HSR

3×

wall

−5/−80

collect

α inv-U

3>6

+62%

rel

3 noises

demo/FM/DR

task-gate

ACT warning

≠ forks

TENTATIVE

Close — Visionary

Rehearse · gate · don't overclaim · no Discord/Drive

1. DAgger queries mistakes; DART rehearses them; 2026 schedules rehearsals.

2. Steal recovery-demo + α -schedule beside $\pi_{0.5}/N1.7$ — not the 2017 stack.

3. Three noises, task-gated α , complementary chunking.

4. TENTATIVE rhyme $\neq$ lineage. ACT is CLEAR. No Discord/Drive checklist.